

23rd International Conference on Knowledge-Based and Intelligent Information & Engineering Systems

Hybrid System for Information Extraction from Social Media Text: Drug Abuse Case Study

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Abstract

Social media are becoming widely used in the healthcare field as a patients-caregivers communication tool giving birth to new sources of information rich with the knowledge that may improve this field. Therefore, social media data analysis becomes a real business requirement for healthcare industrials and data scientists.

However, regarding their complexity and unstructured character, existing natural language processing tools cannot succeed their exploitation. In the literature, a wide range of approaches appeared based on dictionaries, linguistic patterns and machine learning having their strengths and weaknesses.

In this work, we propose a hybrid system combining the above approaches by taking the advantage of each of them to extract structured and salient drug abuse information from health-related tweets. We improve the system accuracy by real time update of the domain dictionary. We collected 1000000 tweets and we conducted different experiments showing the advantage of hybridization on efficient information extraction from social media data.

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Peer-review under responsibility of KES International.

Keywords: Social Media; Natural Language Processing; Classification; Hybrid Approach; ODIN; Information Extraction; Twitter;

1. Introduction

In the healthcare field, social media becomes an important patients-patients and patients-caregivers communication tool and a refuge for many patients especially who suffer from diseases about which they cannot talk to their entourage like substance abuse. Drug abuse is a social phenomenon and a serious health care problem, which is becoming, quickly, an epidemic. It needs quick help regarding the harm it causes to addicted individuals and the society as well. However, traditional solutions are not successful anymore and drug abuse victims are increasingly growing.

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On the other side, drug abusers are using social media such as Twitter, Facebook and drug forums as refuge where they can talk freely about their substance, suffer their problems, make friends and even supply their substances. This phenomenon returns social media into a rich source of clinical data in this field which exploitation may uncover interesting knowledge necessary to make important decisions and reactions to solve drug abuse field problems like overdose prevention, crimes reduction, saving lives, helping abusers reintegrate social life, and why not, stop the dissemination of this poison.

Nevertheless, salient information extraction from health related social text is a tedious task because of their complex character compared to well-formed text sources like EMR-Electronic Medical Records repositories. Social media are unstructured, conversational, informal, short, grammatically and syntactically poor, plenty with colloquial words, abbreviations and slang terms. These problems raise new challenges in the field of information extraction and natural language processing.

In literature, systems based on linguistic rules and others based on machine learning techniques were proposed. However, each of which have their strengths and weaknesses. Indeed, rule-based approaches lack a standard rule language and learning-based approaches require annotated corpus and selective features. Recently, hybrid systems appeared taking advantage of both of them, but, they are not yet used for social data exploitation.

In this paper, we propose a hybrid system for social text analysis in order to extract accurate information namely entities and events. We combine a dictionary-based NLP with linguistic rules and classification to extract accurate information with less labor and less human intervention.

In fact, NLP tools can extract only common named entities like date, number and location, and cannot recognize domain-specific entities like drugs and effects. Thus, we propose to upgrade Stanford Core NLP with an annotator capable to annotate drug names and reactions. The result is an automatically annotated corpus and a set of features to be used in classification. This work requires some human labor in the beginning but once started the process becomes fully automatic. The accuracy of the system is improved by the use of a weighting technique for real time dictionary update with new terms in order to always detect the reliable information from continuously changing sources like social media. In the experiments, we test the efficiency of each method separately on a set of 1000000 tweets collected from Twitter API-Application Program Interface. We tested also the combination of system modules two by two and three by three to find the best combination and best hybridization. We focused also on improving classification model by testing with different features and their combinations in order to keep only most discriminative ones. Final results show that combination of all modules give accurate information and better results.

This paper is organized as follows; in section two we investigate literature works interested into information extraction from social media. In section three, we present the general architecture of the proposed system. Preprocessing is an important step for social text exploitation and presented in section four. In section five, we explain the rule-based approach. Section six is reserved to detail classification approach. In section seven we presented implementation details and obtained results. Finally, we conclude in section eight.

2. Related Work

Information extraction from social media is an ubiquitous problem and still in its infancy. The use of patterns for information extraction started with lexicon-based approaches for named entity recognition using string lookup on domain dictionary or gazetteer like medical ontologies, thesaurus and online clinical data bases [2, 3, 4, 5]. For relation extraction, the use of term matching and word coexistence in the same sentence was not so evident to discover semantic relations. With the development of natural language processing tools, relation extraction becomes much easy thanks to syntactic dependencies. Some approaches make use of lexico-syntactic patterns and others focus on semantic aspects and meaningful relations using lexico-semantic patterns composed of RHS, LHS and predicates. Both methods are closely based on linguistic preprocessing and morphological analysis to find possible relationships between grammatical terms like verbs, nouns and prepositions. [6, 7, 8].

Nevertheless, rule-based information extraction methods suffer from the lack of standard rule language. Moreover, their writing and maintenance require linguistic experts. Despite their efficiency in discovering semantically rich information, these limitations motivate many researchers to adopt data mining techniques. These latter need large and sufficient sets of manually annotated data for more accurate tags prediction of encountered entities, as well as efficient parameters. SVM, HMM, ME and CRF have shown their performance in the identification and prediction of diseases

through detection of symptoms and treatments [9, 10, 11, 12, 13, 14, 15, 16, 17]. The choice of classifier, the selection of features and building the training data set are major difficulties encountered to build a classification model using supervised learning. Moreover, due to misspelling and non-standard abbreviation problems, it is not evident to collect a sufficient training set for clinical term training.

Unsupervised learning is the solution. Clustering like Bayesian model, is used when classes of events are unknown. Each cluster corresponds to an event type [18, 19, 20]. Some approaches use weak supervision [20] that allows information analysts to extract any type of new events by giving only tens of seed examples of the sought event without building an information extractor from scratch. Distant supervision [29] is also used by some contributors where knowledge bases are exploited to assign an entity-level relation label for each pair in texts, and then, they use patterns for characterizing semantic relations. Some authors combined different learning techniques to achieve better results and improve performance [22, 21, 23, 24]. In fact, classification is the most popular task in this field. It requires large sets of manually annotated data. Nevertheless, this is a hard labor and time consuming. Moreover, social data are informal and its vocabulary is still changing and plenty with slangs, so, it is not evident to collect the sufficient training set to learn all possible instances. Thus, a stable and long resisting solution in changing environment is not easily feasible. Clustering is efficient for this task especially to generate over time new classes of events. However, the generated model is a black box difficult to interpret and maintain.

Recent efforts are invested to take advantage of linguistic rules and learning techniques by combining both approaches to succeed information extraction with less limitations and high performance which gave birth to hybrid approaches [25, 26, 27, 28, 35]. However, despite the high performance they achieve compared to machine learning and rule-based approaches individually, hybrid methods are not yet applied for social media data analysis. In this paper, we handle this challenge for Twitter in particular.

3. Proposed Framework

The proposed system starts with a keyword-based crawler developed to collect real time or streaming social media text in particular. After data crawling, data cleaning and linguistic pre-processing, we extract domain specific entities using a customized annotator upgraded to Stanford Core NLP tool based on domain dictionary. The set of entities are promoted to events extraction module which is composed of two components. The first one is based on grammar rules to find events that match the formulated patterns. We obtain an automatically annotated corpus and features useful to conduct a learning process for definitive and accurate events extraction. However, in order to build a continuously up to date system, capable to detect efficiently all events from continuously changing data sources and free vocabulary, we add a third intermediate layer for real time automatic dictionary update. Fig 1 represents an overview of the system and we present detailed description of each layer in subsequent sections.

4. Data Preprocessing

Social text is noisy and requires a cleaning process followed by a complete linguistic preprocessing like POS tagging, lemmatization, common and domain-specific entity extraction.

4.1. Data Cleaning

Social media data are conversational short pieces of text, ungrammatical and informal containing undesirable characters, abbreviations, external links and colloquial words. The straightforward application of existing NLP tools for their processing leads to erroneous and meaningless results. To facilitate further NLP tasks, we develop a cleaning module to remove duplicate data, repeated letters to reduce the word to its canonical form (mariiiijuaaaaa! marijuana), special characters, URLs and URIs, undesirable punctuation and characters (?, !, ;,), etc. After data cleaning, it is easy to apply any NLP tool for linguistic preprocessing.

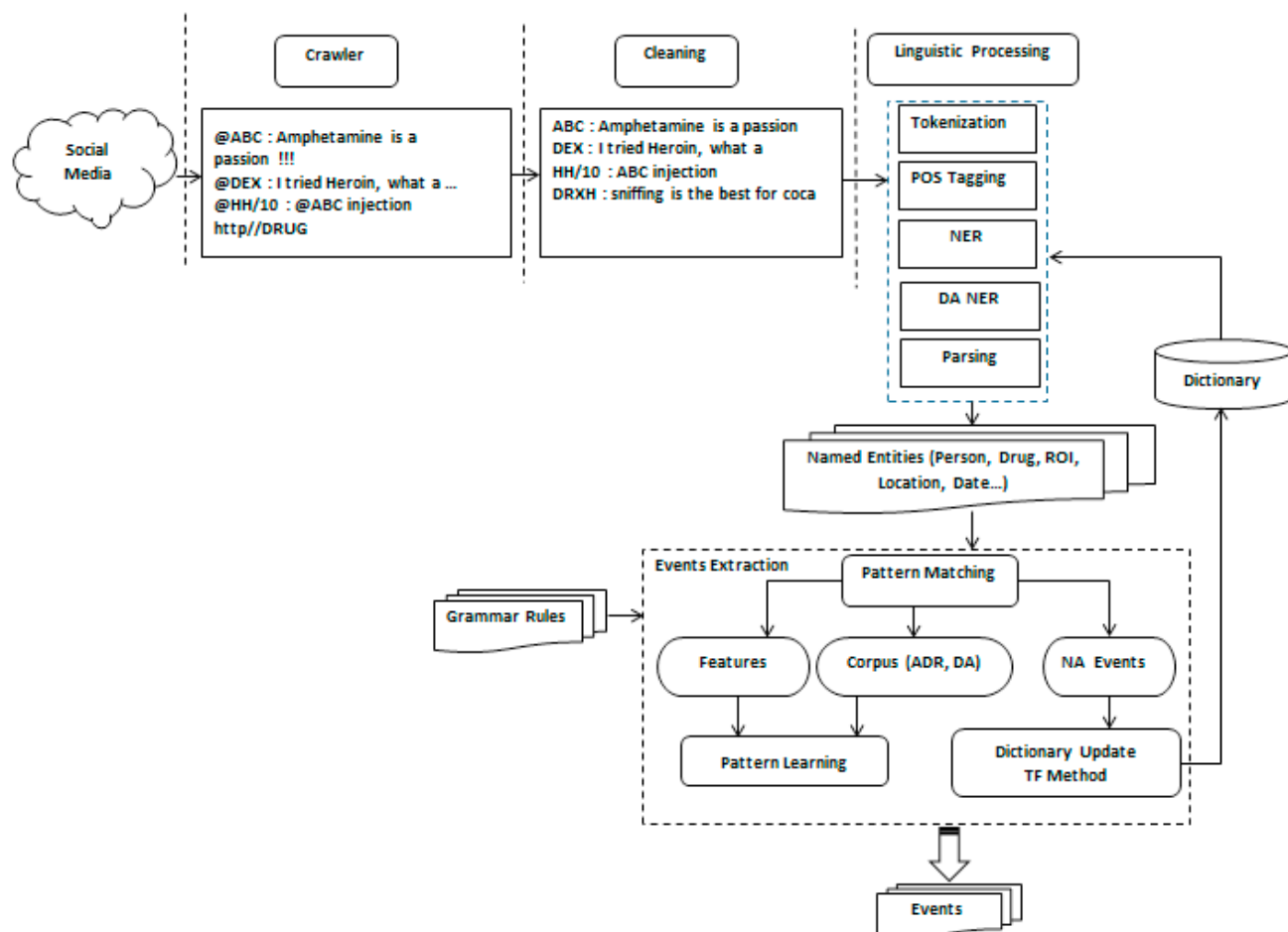


Fig. 1. System Framework Architecture and Methods for Structured Information Extraction from Social Media Text

4.2. Linguistic Preprocessing

We use Stanford CoreNLP [33] which is an open source natural language processing tool that offers the overall pipeline of natural text processing from tokenization, through co-reference resolution, to parsing. For our data, we perform sentence splitting, word tokenization, stemming and lemmatization as well as POS tagging. Named entity recognition-NER is also an important step in the NLP pipeline. It addresses the problem of identification of concepts in free-form text and their classification on predefined types such as Organization, Person, Location, Date, etc. It can additionally include extracting descriptive information from the text about the detected entities through filling of a small-scale template [1]. However, the extraction of domain-dependent entities is hard for standard NLP tools. To solve this problem, we extended Stanford Core NLP capabilities with a customized annotator based on domain dictionary lookup. We use a fuzzy string matching with terms in the dictionary to overcome social data informality; misspellings, non standard abbreviations and slang terms. The proposed annotator is capable to detect six domain entities with their corresponding slangs namely; DRUG (cocaine, heroin, LSD, amphetamine), ROI-Route of Intake (snort, inject, sniff, etc), UNIT (mg, cigarette, ml, etc); PSY-EFF Psychological Effects (depression), PHY-EFF Physical Effects (Cephalegia, chronic pain, etc), MED-COND Medical Condition (overdose). In addition to common named

entities extracted by the standard Stanford NER like Person, Location, Date, Number, etc. The proposed dictionary is fed with seeds entities and their possible slangs. For space and time identification, locations are straightforward detected if they are explicitly mentioned in the text. Otherwise, they are detected from GPS coordinates resolution of the message. So, devices used with users should be equipped with GPS sensors and activated the moment of diffusing posts. Temporal information is also attached to the message and corresponds to the date time of its diffusion. However, sometimes temporal expressions in a message cancel this timestamp which requires another treatment for its identification. For example, the expression 'Yesterday' in the following text: "Yesterday, I vomited all the night because of heroin injection" needs a resolution in order to obtain the correct and valid date of the event. For this end, we use SUTime [30] integrated with Stanford Core NLP which is capable to map any temporal expression to a valid date by reference to the date of the post. When the message doesn't contain any temporal expression that refers to the date of the event, the date of the post is considered by default.

5. Pattern Matching

We focus on events extraction since it is the most complete form of information consisting on named entities and semantic relationships among them. We distinguish simple events and composite events. Regarding the complexity and particularity of social text, a preliminary human effort on information identification is required. Therefore, we use grammar rules to find corresponding linguistic patterns. We make exploitation of ODIN Open Domain Informer [31, 32], a grammar developed for events extraction from biomedical text. It is flexible enough to capture different kinds of events from well-formed or informal text by offering a large set of functionalities and options. Thanks to ODIN and Stanford dependencies [34] we can formulate general patterns capable to detect meaningful relationships between semantic concepts extracted from an external NER system. Semantic concepts are mainly arguments and triggers:

1. Triggers: They are verbs raising the event. We use a semantic class to capture different instances of the trigger (ROI). We define a trigger with the constraint [entity = ROI] or [mention = ROI] and the system looks for the encountered word in the set of instances of the class of entity mentioned in brackets. For adverse drug reaction event types which are expressed with accusative verb (caused, lead, prompt, because of, etc), the corresponding trigger is defined by the lemma of possible verbs.
2. Arguments: They are named entities either built on the fly the moment of writing the pattern, or extracted by an external named entity recognition system.

5.1. Real Time Dictionary Update

In fact, social network conversations between drug abusers had a continuously changing vocabulary. Thus, new slangs, new abbreviations and even new types of drugs as well as side effects may appear through the time, so the system cannot accurately extract all possible relations. To overcome this problem and keep our solution scalable and still up to date capable to detect drug abuse patterns whatever the vocabulary used at any time, we propose a method based on term frequency and weighting technique called TF method to update the proposed dictionary and learn new terms eventually new events. Indeed, not assigned events are the set of named entities missing either triggers or mandatory arguments. The set of sentences that don't create any event mention called S' hereafter, are reduced to vectors of named entities with no semantic relationship between them. Thus, no trigger and consequently no event mention: $S' = \{s_1, s_2, \dots, s_n\}$ where $s_i: [E_1, E_2, \dots, E_n]$.

- Candidate Events Selection: An event mention is not created when one of the mandatory arguments is not matched. So, we keep only sentences containing at least one mandatory argument to be a set of candidate patterns C_p .
- Weights Computation: We use a weighting technique by assigning to each argument or trigger captured in the sentence a weight. The score of the sentence W_s is the normalized sum of the weights of its trigger W_t and n arguments W_{ai} , where W_{ai} is the weight of the argument ai and calculated as follows:
 W_t is the weight of the trigger t . W_t is a fixed weight and should be greater or equal to the weight assigned to other arguments. The obtained scores are greater or equal to 0 and strictly less than 1. Sentences having a null score are removed from C_p . Sentences having a score >0 contain at least one term that belongs to the

$$W_s = \sum_{i=1}^n W_{ai} + W_t \quad (1)$$

$$W_{ai} = \frac{1 - W_t}{\text{number of mandatory arguments}} \quad (2)$$

dictionary. However, to have more chance to capture meaningful events, we should have a maximum number of arguments. Therefore, we retain sentences having only a score ≥ 0.5 . For example; the following sentence "Slalom cocaine" could have a score = 0.5 if we assume that the trigger (slalom in the sentence doesn't belong to the dictionary) has a fixed weight equal to 0.5 and the patterns of interest had only one mandatory argument which is the name of the drug recognized in the sentence and assigned the weight $(1 - 0.5/1 = 0.5)$. So the score of the sentence is: $S = 0.5 + 0 = 0.5$.

- **Relation Mention Detection:** In the set C_p , we look for the co-occurrence of n-grams in sentences such as the bi-gram [slalom cocaine]. 'Slalom' may be a new route of administration of cocaine or any other drug. So, the mentioned bigram is a candidate pattern and 'slalom' is a new term to be added to the dictionary. To achieve this task, we use ODIN dependency pattern to discover a relation mention between terms. A relation mention is a signal of new semantic relationship between entities where one of them is a known concept. It doesn't require any trigger specification. We have just to specify named arguments and the rule is charged of detecting possible relationship between them by following the defined dependency path. This is the process of detecting new entities to update the dictionary. Eventually, future patterns containing these new terms are easily recognized by the proposed system. We use results of Stanford POS tagger to discover missing arguments and triggers as well. For example; we suppose that an abuse event has two mandatory arguments; the trigger which is the route of intake (verb) and the drug name which is a mandatory argument (noun).
- **Term Frequency Computation:** For each captured relation, we compute the frequency of the new term. We propose to use a vector of new terms (t_i), their labels (l_i) and their corresponding frequency (f_i): $V <(t_1, l_1, f_1), \dots, (t_n, l_n, f_n)>$ where (f_i) is the support of the word (number of posts in which it appeared) by the total number of posts processed: $f = \text{support} / \text{NBposts}$.
- **Dictionary Update:** We handle two questions; the first is which term to add? In fact, terms to be added are those having a frequency closest to 1. In fact, more the value is close to 1, more accurate is the information to add to support further events extraction. The second question is how to decide about its nature either it is a slang term or a correct term? We consider a given threshold length of the term. In our experiments we consider words which length is lower than 5, a slang term, else it is a correct term. The added term is removed from the vector V . Eventually, future patterns containing the new terms are recognized by our system as valid patterns, and new local vocabulary is detected automatically without human intervention in the knowledge engineering process.

6. Pattern Learning

We propose to use classification to learn linguistic rules using different set of features to facilitate the process and improve information extraction results as well. The majority of features used in the learning process are inspired from ODIN specification. They are mainly the tag of named entities and some triggers, lexical features for triggers that cannot be represented with a generic named entity, semantic links between named entities which are the semantic description of a complex syntactic path, and the event type. The other features are added to face social data particularity like shortness. In fact, the use of colloquial words and the use of metaphors which may give erroneous results. Minimal events expressed with short sentences and one keyword are not detected with the rule-based module (NA tag) like "marijuana, oh, good experience". For this type of events, we use the sentence length for sentences containing keywords and don't have a complete and correct syntactic structure. Consequently, they are easily classified as

positive instances so valid events. The problem of negations is also treated. Indeed, the linguistic patterns formulated can capture explicit negations in sentences like "I don't use any substance". However, the sentence "like I am under alcohol" is certainly matched to drug abuse patterns (DA tag). However, it is filtered by the classifier thanks to the word window feature type which is very efficient in improving the event detection results. In fact, we propose to use the word before subject (WBS) like in the above example and the word after subject or before trigger (WBT). Table 1 summarizes the set of depicted classification features. Concerning the choice of the adequate learning technique, we trained a set of classifiers and we chose the one achieving best results. Details are presented in the implementation section.

7. Implementation, Experiments and Discussion of Results

For experiments, we chose to analyze Twitter data which are freely and easily accessible from its public Streaming API, unlike other types like Facebook data which could be collected only from specific sites or user profiles. We used the crawler developed at the heart of our system to collect 1000000 tweets according to specific keywords related to the abuse and addiction terminology namely drug names (heroin, cocaine, marijuana, LSD, etc) and effects (overdose, depression). Actually, we didn't use many keywords to detect adverse drug reactions and they are referenced automatically by messages talking about abuse which contain information about suffered reactions. We tried to find the best combination of developed modules to build the best hybrid system for information extraction from Twitter. For each combination, we calculate performance metrics namely precision, recall and F-measure. Results are presented in table 2. The precision is a measure of trust that events detected as positive instances, are really positive. The recall is a measure of trust that all positive instances corresponding to valid events are detected. The F-measure is the harmonic mean between precision and recall representing a trade-of between them. For learning module, the evaluations are based on varying learning features and results are depicted in table 2 as well:

7.1. Named Entity Recognition Results

The customized annotator was implemented using Java eclipse development Environment. We use Apache Commons package which implements Jaro-Winkler distance for fuzzy matching of words encountered in the social text with the words in the dictionary. The system achieved 0.71 of precision, 0.96 of recall and 0.82 of F-measure for a threshold distance fixed to 0.89 and a set of 100 000 tweets. The results are stable when we increase the test data set. When tested for 1000000 tweets the accuracy parameters were confirmed which show the efficiency and stability of the proposed solution. It is able to recognize names of drugs, psychological effects, physical effects, medical conditions, units and routes of intake with a very interesting recall. In fact, the precision is a measure of trust that instances marked as positives are really positive. However, the recall is a measure of trust that all positive instances are identified. The annotator achieved higher recall to which we are interested since we are handling social media text in which not all positive instances are written correctly.

7.2. Experiment 1

For the first test, we used only the rule-based approach to extract events. When tested for the 100000 tweets, it achieved 0.53 of precision, 0.74 recall and 0.61 of F-measure. The performance of the rule-based approach individually is poor since it generates a large number of false positives as well as false negatives. Moreover, the performance measures decrease when we enlarge the test data set. In fact, many relevant events are missed and invalid events are tagged. In addition, this module is built on the basis of a dictionary-based named entity recognizer which despite the interesting results it achieved, it cannot be accurate for longer, so need to be updated with new terms in order to extract all events and increase system accuracy.

7.3. Experiment 2

To improve system accuracy, we implemented a module for automatic and real time dictionary update based on failures of the rule based module. Indeed, sentences and terms which are not tagged by the named entity recognition

Table 1. Classification Features for Pattern Learning.

Feature Type	Description	Values
Named Entities (NE)	The tag of named entity which could be an argument or a trigger. They are binary features.	PATIENT, DRUG, NUMBER, UNIT, ROI, MANNER
Lexical features (LF)	The lemma of verbs representing some triggers that don't correspond to any named entity	EFFECT, PHY-EFF, PSY-EFF "cause", "have", "take", "be" "prompt", "feel", "test", "get", "do" "want" "love", "like", "purchase" "hate" "need", "play", "give"
Semantic Links (SL)	Semantic relationships between the trigger and the argument	Semantic relationships between the trigger and the argument
Event Type	the event tag	DA, ADR, NA
Sentence Length	Number of non stop words in the sentence	five, lfive, gfive
WBS	The lexical match of the word before subject. NA if it doesn't exist	Like, if, can, would, could, NA
WBT	The lexical match of the word before trigger. NA if it doesn't exist	Should, safely, NA

Table 2. The results obtained by each module for 1000000 tweets. RBM is the rule-based method. TF is the term frequency method for real time dictionary update. LBM is the learning-based method. We tried to combine each time a set of modules and calculate performance measures to find the best combination for the target hybrid system architecture.

Experiment		Drug Abuse Event			Adverse Drug Reaction		
		P	R	F	P	R	F
1	RBM	0.43	0.68	0.52	0.45	0.71	0.55
2	RBM/TF	0.61	0.72	0.58	0.67	0.79	0.65
3	RBM/LBM(all features)	0.93	0.93	0.93	0.98	0.97	0.97
4	RBM/TF/LBM(All Features)	0.95	0.95	0.95	0.98	0.98	0.96
5	RBM/ TF/ LBM (NE, LF, SL)	0.75	0.77	0.74	0.78	0.83	0.75

Table 3. Classification results for Drug abuse and adverse drug reaction events.

	Drug Abuse Event			Adverse Drug Reaction		
	P	R	F	P	R	F
Decision Tree (J48)	0.93	0.93	0.93	0.98	0.97	0.97
Logistic Regression	0.91	0.9	0.9	0.98	0.97	0.97
SVM (Libsvm)	0.93	0.93	0.93	0.98	0.97	0.97
Naïve Bayes	0.84	0.83	0.83	0.96	0.95	0.95

and relations which are not detected by the rule based system are either not potential events or new terms. In this experiment, we observed the results obtained by combining the named entity recognition, rule-based module and the automatic dictionary update layer. We want to calculate accuracy measures when we combine these two methods. As depicted in the table 2, we achieved 0.61 of precision which is better than the results of the rule-based system individually. Indeed, thanks to the automatic dictionary update module we can detect all the time new terms, add them to the dictionary and extract all possible events.

7.4. Experiment 3

Due to the informality of social text, we define the majority, if not all, the arguments optional in their corresponding rules in order to capture all possible positive instances (events) which bias the results and leads to an increasing number of false positives and false negatives. Therefore, we propose to build a classification model capable to extract

meaningful events and distinguish real positive instances from false positives in order to increase the precision of the system. The process of information extraction is transformed into a pattern learning process based on linguistic layer output. Machine learning algorithms offer larger possibilities to use all relevant features necessary to extract the most reliable information. The choice of the learning technique is the focus of this study. Depicted features are considered binary variables which values are 'true' if they exist in the event and 'false' if not. For each type of event, we have an important set of features with 'false' values since they belong to the other type, which bias the results and gives poor performance. Therefore, we build two classifiers trained separately each one with its own relevant features. We divided the set of features into features for detecting drug abuse events and features relevant to detect adverse reaction events. Thus, we obtained two classification models which work in parallel on the same data generated by the pattern matching module. We trained four classification algorithms namely J48 for Decision Tree, Logistic Regression, Libsvm for Support Vector Machine and Naive Bayes implemented in Weka. We divided the training set into 0.60 for training and 0.40 for test. Table 3 summarizes obtained results for each type of events and with each learning technique. Naive Bayes performs badly for both types for events and Logistic Regression is not the adequate classifier of drug abuse events. Thus, the classifier to integrate with the system pipeline is either the decision tree or support vector machine regarding the good results they achieve. Events obtained from the rule matching module passed through the learning layer using a set of features depicted in the table above in section 4 achieved a best precision of 0.93. However, without dictionary update, the system will be static and obtained results cannot be trusted for longer. This will affect the whole system accuracy and its performance will decrease through the time.

7.5. Experiment 4

Seeking for better performance, we integrated the real time dictionary update process to running solution. We tested the system with all modules implemented together and learning features. We achieved 0.95 of precision for drug abuse events and 0.98 to extract Adverse Drug Reactions. This experiment shows the best scenario. In fact, the use of a rule-based approach generates a set of efficient classification features and promotes all the time an automatically annotated training set to the learning-based component so we don't require additional annotation efforts for each new type of event. We improved all weaknesses of our system by the use of real time and automatic update of the domain dictionary which give the system the capability of continuously extracting accurate events.

7.6. Experiment 5

By including the automatic dictionary update module, we proposed to remove some features in order to improve processing time and save storage capacity. We tested the system using only three types of features namely; named entity (NE), semantic links (SL) and lexical features (LF). We observe in table 2 that performance measures such as precision decreases when we remove the features: event type, sentence length, word before trigger and word before subject.

8. Conclusion

Along this work, we tried to prove that the use of machine learning and linguistic rules separately is not enough to achieve better results for information extraction from social media. The use of a linguistic method based on dictionary without real time dictionary update is also a failed solution. So, we combined linguistic rules, machine learning and domain dictionary to extract meaningful, complete, structured and useful information called event by achieving good performance. However, the simple implementation of our system in a large scale and evolvable environment like big data will not resist for long. Therefore, the use of parallel and distributed architecture is a necessity to face big data variety, volume and velocity which we will handle in future work.

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